

Monte Carlo Coulomb Collisions for Guiding-Centre Particles in Boozer Coordinates: Implementation Notes

firm3d — ep_thermal_collisions

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1 Overview

These notes describe the algorithm used in `trace_particles_boozer_with_collisions` to include stochastic Coulomb collisions for an energetic particle (EP) tracing through one or more Maxwellian background species.

The approach couples the deterministic guiding-centre (GC) orbit equations in Boozer coordinates with a Milstein stochastic term for the velocity-space diffusion. The full state vector is $(s, \theta, \zeta, v, \xi)$, where $v = |\mathbf{v}|$ is the total speed and $\xi = v_{\parallel}/v$ is the pitch.

Primary references:

- Hirvijoki et al. [1] — Monte Carlo implementation of the guiding-centre Fokker–Planck equation in (v, ξ) coordinates; Eqs. (32)–(33) of that paper define the SDE system used here.
- Hirvijoki et al. [2] — ASCOT4/5 code description; source file `mccc_coefs.h` defines the Coulomb logarithm convention adopted here.
- Boozer & Kuo-Petravic [4] — pitch-angle representation in Boozer coordinates.

2 Guiding-Centre Equations in Boozer Coordinates

In Boozer coordinates (s, θ, ζ) , where s is the normalised toroidal flux, θ the poloidal angle, and ζ the toroidal angle, the vacuum GC equations are

$$\dot{s} = -\frac{m v^2 (1 + \xi^2)}{2 q \psi_0} \frac{\partial B}{\partial \theta} \frac{1}{B}, \quad (1)$$

$$\dot{\theta} = \frac{m v^2 (1 + \xi^2)}{2 q \psi_0} \frac{\partial B}{\partial s} \frac{1}{B} + \frac{\iota \xi v B}{G}, \quad (2)$$

$$\dot{\zeta} = \frac{\xi v B}{G}, \quad (3)$$

$$\dot{\xi} = -\frac{(\iota \partial_{\theta} B + \partial_{\zeta} B) v (1 - \xi^2)}{2G} \quad [\text{mirror force}], \quad (4)$$

where ψ_0 is the reference poloidal flux, $G(s)$ is the covariant toroidal field component, and $\iota(s)$ is the rotational transform. The magnetic moment $\mu = v^2(1 - \xi^2)/(2B)$ is reconstructed locally at each evaluation.

With collisions the equations for v and ξ acquire deterministic drift terms (Section 4) and stochastic noise terms (Section 5.2).

3 Background Species and Profiles

Each background species b is described by a `ThermalBackground` object specifying:

- $n_b(s)$ — number density [m^{-3}], callable of the flux label $s \in [0, 1]$;
- $T_b(s)$ — temperature [J], callable (use `T_keV × 1e3 × e`);

- m_b — species mass [kg];
- q_b — species charge [C] (signed).

Internally the profiles are pre-evaluated on a uniform s -grid of 512 points and linearly interpolated in C++.

4 Collision Coefficients

Notation

The symbols used here follow the ASCOT5 source code (`mccc_coefs.h`, [3]); Table 1 shows the correspondence with Hirvijoki et al. [1], whose Eqs. (32)–(33) define the SDE structure.

This paper	ASCOT5 <code>mccc_coefs.h</code>	Hirvijoki et al. [1]
$D_{\parallel}$	<code>Dpara</code>	$D_k/2$
ν_D	<code>nu</code> = $2D_{\perp}/v^2$	$2D_{\perp}/v^2$
Q	<code>Q</code>	(Einstein-relation drag in K)
K	<code>K</code> = $Q + D'_{\parallel} + 2D_{\parallel}/v$	K (drift in Eq. 32)

Table 1: Notation correspondence. The noise amplitude for v is $g_v = \sqrt{2D_{\parallel}}$ (this paper) = $\sqrt{D_k}$ (Hirvijoki 2013).

4.1 Debye Length

The total Debye screening length from all background species is [5, 6]

$$\frac{1}{\lambda_D^2} = \sum_b \frac{n_b q_b^2}{\varepsilon_0 T_b}. \quad (5)$$

4.2 Coulomb Logarithm

The Coulomb logarithm is defined as the ratio of maximum to minimum impact parameters [5, 6, 7]:

$$\ln \Lambda_{ab} = \ln \left(\frac{\lambda_D}{b_{\min}} \right), \quad b_{\min} = \max(b_{\text{cl}}, b_{\text{qm}}). \quad (6)$$

The maximum impact parameter is the Debye length (Eq. (5)). The minimum impact parameter is the larger of the classical 90°-deflection distance and the quantum-mechanical de Broglie wavelength of the reduced mass:

$$b_{\text{cl}} = \frac{|q_a q_b|}{4\pi\varepsilon_0 m_r \bar{v}^2}, \quad (7)$$

$$b_{\text{qm}} = \frac{\hbar}{2m_r \sqrt{\bar{v}^2}}, \quad (8)$$

where $m_r = m_a m_b / (m_a + m_b)$ is the reduced mass.

The effective velocity scale $\bar{v}^2$ is taken as

$$\bar{v}^2 = v^2 + v_{\text{th},b}^2, \quad v_{\text{th},b} = \sqrt{2T_b/m_b}, \quad (9)$$

following the ASCOT5 source convention [3]. This form is a numerical interpolation device, not derived from first principles: Ichimaru [7] derives b_{cl} and b_{qm} only in the fast-particle limit $v \gg v_{\text{th},b}$ (Section 4.3B of that reference), where $\bar{v} \approx v$. The $\bar{v}^2$ form extends the formula to the slow-particle regime without a discontinuous switch: fast EP against ions ($v \gg v_{\text{th},b}$, $\bar{v} \approx v$) and slow EP against electrons ($v \ll v_{\text{th},e}$, $\bar{v} \approx v_{\text{th},e}$).

The quantum branch $b_{\text{qm}} > b_{\text{cl}}$ applies when

$$\frac{2\pi\epsilon_0 \hbar \sqrt{\bar{v}^2}}{|q_a q_b|} > 1, \quad (10)$$

i.e. when the de Broglie wavelength of the reduced mass particle exceeds the classical distance of closest approach. For 3.52 MeV α -particles scattering off 10 keV electrons one finds $b_{\text{qm}} \approx 22 b_{\text{cl}}$, so the quantum branch is active throughout alpha slowing-down against electrons. A warning is printed if $\ln \Lambda < 2$, where the perturbative binary-collision expansion underlying the Fokker-Planck operator begins to lose validity.

4.3 Chandrasekhar G Function

The Chandrasekhar function [8] is

$$G(x) = \frac{\text{erf}(x) - \frac{2x}{\sqrt{\pi}} e^{-x^2}}{2x^2}, \quad x = v/v_{\text{th},b}. \quad (11)$$

Its derivative satisfies the recurrence

$$G'(x) = \frac{2}{\sqrt{\pi}} e^{-x^2} - \frac{2G(x)}{x}. \quad (12)$$

Key limiting behaviours:

$$\begin{aligned} x \rightarrow 0: \quad G &\rightarrow 0, \quad G' \rightarrow 2/(3\sqrt{\pi}) \approx 0.376; \\ x \rightarrow \infty: \quad G &\rightarrow 1/(2x^2), \quad G' \rightarrow 0. \end{aligned}$$

4.4 Collision Frequency Prefactor

Define the Rosenbluth prefactor for species b [9, 10]:

$$\Gamma_b = \frac{n_b q_a^2 q_b^2 \ln \Lambda_{ab}}{4\pi\epsilon_0^2 m_a^2}. \quad (13)$$

4.5 Pitch-Angle Scattering Frequency

$$\nu_D^{(b)} = \frac{\Gamma_b [\text{erf}(x) - G(x)]}{v^3} \equiv \frac{2D_{\perp}^{(b)}}{v^2}. \quad (14)$$

The second form uses the perpendicular diffusion coefficient $D_{\perp}^{(b)} = \Gamma_b [\text{erf}(x) - G(x)]/(2v)$ and matches ASCOT5's `mccc_coefs_nu` and the $2D_{\perp}/v^2$ notation of Hirvijoki et al. [1, 2].

4.6 Parallel Diffusion Coefficient

$$D_{\parallel}^{(b)} = \frac{\Gamma_b G(x)}{v}, \quad \frac{\partial D_{\parallel}^{(b)}}{\partial v} = \frac{\Gamma_b}{v} \left(\frac{G'(x)}{v_{\text{th},b}} - \frac{G(x)}{v} \right). \quad (15)$$

In Hirvijoki et al. [1] the parallel diffusion coefficient is denoted D_k , with $D_{\parallel} = D_k/2$; the noise amplitude for the speed SDE is $\sqrt{D_k} = \sqrt{2D_{\parallel}}$ in both conventions. ASCOT5 names this coefficient `Dpara`. The v -derivative is required by the Milstein correction term (Section 5.2).

4.7 Deterministic Drag in v

The drag coefficient (ASCOT5: `mccc_coefs_Q` [3]) is fixed by the Einstein relation:

$$Q^{(b)} = -\frac{m_a v}{T_b} D_{\parallel}^{(b)} = -\Gamma_b \frac{m_a G(x)}{T_b}. \quad (16)$$

The total deterministic drift in speed is

$$K^{(b)} = Q^{(b)} + \frac{\partial D_{\parallel}^{(b)}}{\partial v} + \frac{2 D_{\parallel}^{(b)}}{v} \quad (17)$$

(ASCOT5's `mccc_coefs_K` [3]; the drift coefficient K of Eq. (32) in Hirvijoki et al. [1]).

Itô transformation leading to Eq. (17)

Equation (17) can be obtained two independent ways, which cross-check each other.

(i) Itô's formula on $v = |\mathbf{v}|$. In three-dimensional velocity space the collisional motion of the test particle is

$$d\mathbf{v} = F_{\text{dyn}} \hat{\mathbf{v}} dt + \boldsymbol{\sigma} \cdot d\mathbf{W}, \quad \boldsymbol{\sigma} \boldsymbol{\sigma}^{\top} = 2D_{\parallel} \hat{\mathbf{v}} \hat{\mathbf{v}} + 2D_{\perp} (\mathbf{I} - \hat{\mathbf{v}} \hat{\mathbf{v}}), \quad (18)$$

where F_{dyn} is the dynamical friction [8, 10, 11]. The speed $v = |\mathbf{v}|$ is a nonlinear function of $\mathbf{v}$, so Itô's formula applies with $\nabla_{\mathbf{v}} v = \hat{\mathbf{v}}$ and Hessian $\nabla_{\mathbf{v}} \nabla_{\mathbf{v}} v = (\mathbf{I} - \hat{\mathbf{v}} \hat{\mathbf{v}})/v$:

$$dv = \left(F_{\text{dyn}} + \frac{2D_{\perp}}{v} \right) dt + \sqrt{2D_{\parallel}} dW_v. \quad (19)$$

The $2D_{\perp}/v$ drift is the Bessel-process effect: kicks transverse to $\mathbf{v}$ do not change v at first order, but they lengthen the vector at second order, $\Delta v = |\mathbf{v} + \Delta \mathbf{v}_{\perp}| - v \approx |\Delta \mathbf{v}_{\perp}|^2/2v > 0$.

(ii) Fokker–Planck matching (detailed balance). The Itô SDE $dv = K dt + \sqrt{2D_{\parallel}} dW_v$ evolves the speed density $P(v) \propto v^2 f(v)$ according to $\partial_t P = -\partial_v (K P) + \partial_v^2 (D_{\parallel} P)$. Zero flux on the equilibrium $P_{\text{eq}} \propto v^2 e^{-m_a v^2/2T_b}$ requires $K P_{\text{eq}} = \partial_v (D_{\parallel} P_{\text{eq}})$, i.e.

$$K = -\frac{m_a v}{T_b} D_{\parallel} + \frac{\partial D_{\parallel}}{\partial v} + \frac{2 D_{\parallel}}{v}, \quad (20)$$

which is Eq. (17) with Q as in Eq. (16). Here $\partial_v D_{\parallel}$ is the multiplicative-noise (Itô) drift that appears when the diffusion term is commuted into advective form, and $2D_{\parallel}/v$ comes from the Jacobian v^2 of spherical velocity-space coordinates.

Consistency of (i) and (ii). For Maxwellian field particles the dynamical friction obeys the identity

$$F_{\text{dyn}}^{(b)} = -\left(1 + \frac{m_a}{m_b}\right) \frac{2x^2 G(x) \Gamma_b}{v^2} = Q^{(b)} + \frac{\partial D_{\parallel}^{(b)}}{\partial v} + \frac{2(D_{\parallel}^{(b)} - D_{\perp}^{(b)})}{v}, \quad (21)$$

so that $F_{\text{dyn}} + 2D_{\perp}/v = Q + \partial_v D_{\parallel} + 2D_{\parallel}/v = K$ exactly. (F_{dyn} is ASCOT5's `mccc_coefs_F`, which is not used in the guiding-centre scheme.) In the fast-particle limit $x \gg 1$ this identity encodes the classic cancellation

$$F_{\text{dyn}} + \frac{2D_{\perp}}{v} \longrightarrow -\left(1 + \frac{m_a}{m_b}\right) \frac{\Gamma_b}{v^2} + \frac{\Gamma_b}{v^2} = -\frac{m_a}{m_b} \frac{\Gamma_b}{v^2} :$$

most of the dynamical friction is compensated by transverse diffusion, and the net slowing-down rate is set by the mass ratio. Note in particular that Q is *not* the dynamical friction — it is the Einstein-relation drag $-(m_a v/T_b) D_{\parallel}$.

4.8 Summation over Species

All coefficients are additive over background species:

$$\nu_D = \sum_b \nu_D^{(b)}, \quad D_{\parallel} = \sum_b D_{\parallel}^{(b)}, \quad K = \sum_b K^{(b)}. \quad (22)$$

The Debye length used in Eq. (6) is computed from all species simultaneously (Eq. (5)), prior to the per-species loop.

5 Stochastic Differential Equations

5.1 Full SDE System

The complete SDE for the five-dimensional state $(s, \theta, \zeta, v, \xi)$ is [1]

$$ds = \dot{s}_{\text{orb}} dt, \quad (23)$$

$$d\theta = \dot{\theta}_{\text{orb}} dt, \quad (24)$$

$$d\zeta = \dot{\zeta}_{\text{orb}} dt, \quad (25)$$

$$dv = K(v, s) dt + \sqrt{2D_{\parallel}} dW_v, \quad (26)$$

$$d\xi = \dot{\xi}_{\text{mirror}} dt - \xi \nu_D dt + \sqrt{\nu_D(1 - \xi^2)} dW_{\xi}, \quad (27)$$

where dW_v and dW_{ξ} are independent Wiener increments with $\mathbb{E}[dW^2] = dt$. Equations (26)–(27) correspond to Eqs. (32)–(33) of Hirvijoki et al. [1], with the identification $D_{\parallel} \leftrightarrow D_k/2$ and $\nu_D \leftrightarrow 2D_{\perp}/v^2$ in their notation. The noise amplitude for ξ ensures that the marginal distribution of ξ converges to a uniform distribution on $[-1, 1]$ in the isotropisation limit $\nu_D t \gg 1$.

5.2 Milstein Scheme

After each accepted Dormand–Prince (DP) step of physical size h , the stochastic increments $\Delta W_v \sim \mathcal{N}(0, h)$ and $\Delta W_{\xi} \sim \mathcal{N}(0, h)$ are generated and applied via the Milstein method [12]:

$$v \leftarrow v + g_v \Delta W_v + \frac{1}{2} \frac{\partial g_v}{\partial v} (\Delta W_v^2 - h), \quad (28)$$

$$\xi \leftarrow \xi + g_\xi \Delta W_\xi + \frac{1}{2} \frac{\partial g_\xi}{\partial \xi} (\Delta W_\xi^2 - h), \quad (29)$$

where the noise amplitudes and their derivatives are

$$g_v = \sqrt{2D_\parallel}, \quad \frac{\partial g_v}{\partial v} = \frac{\partial D_\parallel / \partial v}{\sqrt{2D_\parallel}}, \quad (30)$$

$$g_\xi = \sqrt{\nu_D(1 - \xi^2)}, \quad \frac{\partial g_\xi}{\partial \xi} = \frac{-\xi \nu_D}{\sqrt{\nu_D(1 - \xi^2)}}. \quad (31)$$

In terms of the coefficient structs, the Milstein correction for ξ simplifies to

$$\delta \xi_{\text{Milstein}} = -\frac{1}{2} \xi \nu_D (\Delta W_\xi^2 - h). \quad (32)$$

Physical bounds are enforced after each step: $v \geq 0$ and $\xi \in [-1, 1]$.

6 Adaptive Integration Strategy

6.1 Dormand–Prince Solver

The deterministic ODE part (all five components of the drift) is integrated with the classic 4(5)-order embedded Dormand–Prince (FSAL) method [13]. Step-size control uses mixed absolute/relative tolerances:

$$\text{err}_i = |y_i^{(5)} - y_i^{(4)}| \leq \delta_{\text{abs}} + \delta_{\text{rel}} |y_i|. \quad (33)$$

Default tolerances: $\delta_{\text{abs}} = \delta_{\text{rel}} = 10^{-9}$.

6.2 Minimum Step Size

The parameter `DP_hmin` (default 0.0) sets a lower bound on the physical step size h in seconds. A value of 10^{-10} – 10^{-9} s prevents the solver from stalling at bounce points while keeping steps small enough that the orbital dynamics remain accurate.

6.3 Operator Splitting

The SDE is solved via Lie–Trotter operator splitting:

1. **Deterministic half:** advance $(s, \theta, \zeta, v, \xi)$ by one DP step of physical time h using the full drift $(\dot{s}, \dot{\theta}, \dot{\zeta}, K, -\xi \nu_D + \dot{\xi}_{\text{mirror}})$.
2. **Stochastic half:** apply Milstein noise to (v, ξ) at the accepted state, using h as the diffusion window.

The trajectory is saved *before* the Milstein step at each t_{save} checkpoint, so saved states correspond to the deterministic prediction (consistent with the standard output convention).

7 Output Format

Each particle trace is a NumPy array of shape (ntimesteps, 6) with columns

$$[t, s, \theta, \zeta, v_{\parallel}, v].$$

From these the kinetic energy $E = \frac{1}{2}mv^2$ and the magnetic moment $\mu = (v^2 - v_{\parallel}^2)/(2B)$ can be reconstructed at each saved point. When `forget_exact_path=True` only the first and last rows are retained.

Each particle uses RNG seed `rng_seed+i`, making multi-particle MPI runs fully reproducible.

8 Python Interface

```
from firm3d.field.collisions import ThermalBackground, \
    trace_particles_boozers_with_collisions
from firm3d.util.constants import (
    PROTON_MASS, ELECTRON_MASS, ELEMENTARY_CHARGE, ONE_EV,
    ALPHA_PARTICLE_MASS, ALPHA_PARTICLE_CHARGE,
    FUSION_ALPHA_PARTICLE_ENERGY,
)

# Typical DT-fusion plasma backgrounds
n0 = 1e20          # m^-3
deuterium = ThermalBackground(
    n_profile=lambda s: n0 * (1 - s),
    T_profile=lambda s: 10e3 * ONE_EV * (1 - s),
    mass=2 * PROTON_MASS,
    charge=ELEMENTARY_CHARGE,
)
tritium = ThermalBackground(
    n_profile=lambda s: n0 * (1 - s),
    T_profile=lambda s: 10e3 * ONE_EV * (1 - s),
    mass=3 * PROTON_MASS,
    charge=ELEMENTARY_CHARGE,
)
electrons = ThermalBackground(
    n_profile=lambda s: 2 * n0 * (1 - s),    # quasi-neutral
    T_profile=lambda s: 10e3 * ONE_EV * (1 - s),
    mass=ELECTRON_MASS,
    charge=-ELEMENTARY_CHARGE,
)

res_tys, res_hits = trace_particles_boozers_with_collisions(
    field,
    stz_inits,                                # (N, 3) array
    parallel_speeds,                          # (N,) array, m/s
    backgrounds=[deuterium, tritium, electrons],
```



```

    tmax=1e-1,                                # 100 ms
    mass=ALPHA_PARTICLE_MASS,
    charge=ALPHA_PARTICLE_CHARGE,
    Ekin=FUSION_ALPHA_PARTICLE_ENERGY,        # 3.52 MeV
    tol=1e-9,
    dt_save=1e-4,                              # save every 0.1 ms
    DP_hmin=1e-10,                             # s
    rng_seed=42,
)

```

9 Validity and Limitations

- **Vacuum field:** electric fields, $E \times B$ drifts, and electromagnetic perturbations are not included in the collision model (though perturbations can be combined via the existing SAW tracing infrastructure).
- **Maxwellian backgrounds:** the collision operator assumes each background species has a local Maxwellian distribution function. Distorted distributions (e.g. beam-driven) are not supported.
- **Test-particle limit:** the EP density is assumed negligibly small compared to the background, so no back-reaction on the background is included.
- **Guiding-centre approximation:** valid when the Larmor radius $r_L = mv_\perp/(|q|B)$ is small compared to the scale lengths of B and the density/temperature profiles. For 3.52 MeV α -particles in a 5 T stellarator $r_L \sim 5$ cm, comparable to device scale, so finite-Larmor-radius corrections may be non-negligible.
- **Zeroth-order GC collision operator:** the implementation follows Hirvijoki et al. [1] at zeroth order in ρ_*/L (Larmor radius over gradient length). The spatial diffusion term $D_X \propto D_\perp/\Omega_g^2$ present in the full operator [2] is neglected.
- **Slow tests:** `@pytest.mark.slow` — five physics-validation tests require $N \sim 20$ particles integrated for 5–100 μ s and are intended for HPC resources (Perlmutter or similar).

References

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